FitBuddy: Model Selection & Architecture Guide

AI Fitness Plan Generator using Gemini Models | Epic 1 Implementation Document

1. Overview of Model Selection & Architecture (Epic 1)

The foundation of ***FitBuddy*** relies on selecting the appropriate Generative AI model and establishing a robust system architecture. This phase ensures that the application can efficiently process user constraints, manage backend routing, and deliver structured fitness regimens securely.

> **Key Milestone Objective:** Research and select Google's Gemini generative model for text processing, define the application architecture, and configure secure API integration workflows.

2. Story 1: Research and Select Appropriate Generative AI Model

- **Model Choice:** Google Gemini Pro (***gemini-pro***) was selected for its exceptional natural language understanding, robust multi-turn reasoning capabilities, and ability to format structured markdown outputs (ideal for fitness and nutrition schedules).
- **API Integration:** Utilizes the official ***google-generativeai*** Python SDK to establish secure, token-authenticated communication with Google's servers.

3. Story 2: Define Application Architecture

The system follows a modern, modular MVC-inspired pattern using FastAPI:

- **Presentation Layer (Frontend):** HTML5, CSS3, and Jinja2 templating for rendering interactive user forms and responsive output dashboards.
- **Application Layer (Backend):** FastAPI server (***app.py***) managing server lifecycle, static assets, and modular routing (***routes.py***).
- **Intelligence Layer (AI Engine):** Prompt-engineering pipeline that translates raw user metrics into contextual fitness prompts processed via the Gemini model.

4. Environment & Security Configuration

To maintain security best practices, API keys are isolated using environment variables (***python-dotenv***), ensuring credentials are never hardcoded or exposed on the client-side frontend.

5. Summary

By establishing a clear architecture and selecting the ***gemini-pro*** model, ***FitBuddy*** ensures high reliability, speed, and safety in generating personalized health plans.